Table 1. Data audit

Rows 1–10 of 10

Measure	Value
Runs	100
Requirements	13
Criteria	7
Possible requirement–criterion checks	9100
Flag explanations	3229
Not-flagged checks	5871
Unique explanation strings	2566
Observed requirement–criterion pairs	54
Labels human-reviewed	0
Explanations with no matched reason	8

The 13 requirements are fixed cases. The repeated runs are the resampling units.

Table 2. Proposed reason definitions

Rows 1–8 of 28

reason_id	label	definition
R01	Missing stated provenance	The explanation says a value, bound, constraint, or necessity lacks a documented need, source, trace, or rationale.
R02	Prescribed design solution	The explanation criticizes imposing an implementation instead of describing the required outcome.
R03	Abstraction or system boundary	The explanation says a detail belongs at another system level or outside the system of interest.
R04	Notes mixed into requirement	The explanation criticizes explanatory notes, assumptions, or scoping material appearing with the requirement.
R05	Multiple obligations	The explanation identifies two or more capabilities, characteristics, or constraints within one requirement.
R06	Indirect capability phrasing	The explanation criticizes the phrase shall be able to.
R07	Negative obligation wording	The explanation criticizes a negative statement such as shall not.
R08	Bound wording criticized	The explanation criticizes no more than, no less than, or not more than phrasing.

Proposed categories are editable. Inclusion/exclusion rules and regexes are supplied in config/reason_codebook.json.

Table 2. Proposed reason definitions

Rows 9–16 of 28

reason_id	label	definition
R09	Dependence on another requirement	The explanation says cross-references or implicit links make the requirement insufficiently self-contained.
R10	Endpoint versus path accuracy	The explanation distinguishes accuracy at the endpoint from accuracy throughout the path.
R11	Unit or numerical equivalence	The explanation alleges inconsistent metric/imperial or other numerical conversions.
R12	Environmental bounds undefined	The explanation says qualitative environmental terms or missing tolerances leave the operating envelope unclear.
R13	Particulate limit or measurement unspecified	The explanation asks for particle size, quantity, concentration, threshold, or measurement conditions.
R14	Leakage or containment undefined	The explanation says enclosure or leakage has no defined containment, sealing, or leak-rate criterion.
R15	Trajectory or motion scope unspecified	The explanation questions the unspecified path, direction, distance, duration, or scope of the required movement.
R16	Payload or interface details omitted	The explanation asks for mass, size, attachment, or interface attributes.

Proposed categories are editable. Inclusion/exclusion rules and regexes are supplied in config/reason_codebook.json.

Table 2. Proposed reason definitions

Rows 17–24 of 28

reason_id	label	definition
R17	Initiating condition omitted	The explanation asks for an electrical signal, trigger, or initiating condition.
R18	Coordinates or reference frame omitted	The explanation asks for coordinates, coordinate frame, origin/destination definition, or a reference frame.
R19	Energy-budget scope unclear	The explanation questions what all operations covers or the duration/cycle basis of the energy limit.
R20	Gravity orientation or test context unclear	The explanation questions the meaning of perpendicular gravity or whether it describes operation, test, or mounting.
R21	Gravity versus mission-context tension	The explanation says normal Earth gravity conflicts with or is not derived from the stated ISS operating context.
R22	Timed interval boundaries missing	The explanation says start/stop events for a timing constraint are undefined.
R23	Accuracy verification conditions missing	The explanation asks for a datum, load, reference point, measurement method, or test condition for accuracy.
R24	Duration consistency questioned	The explanation alleges a conflict or lack of reconciliation between one minute and the movement time limits.

Proposed categories are editable. Inclusion/exclusion rules and regexes are supplied in config/reason_codebook.json.

Table 2. Proposed reason definitions

Rows 25–28 of 28

reason_id	label	definition
R25	Absolute prohibition unverifiable	The explanation questions the feasibility or verification of an absolute zero-particulate constraint.
R26	Implied power constraint	The explanation infers a power level from the stated energy and duration.
R27	Assumed component existence	The explanation questions whether lubricated components exist in the system.
R28	Claimed environment-value mismatch	The explanation claims stated environmental values do not match an external ISS value.

Proposed categories are editable. Inclusion/exclusion rules and regexes are supplied in config/reason_codebook.json.

Table 3. Flag counts out of 100

Rows 1–13 of 13

requirement_id	A2	A3	A4	A5	A6	A9	A10
FR.1	0	4	29	48	58	0	100
FR.2	0	0	0	52	0	0	100
PR.1	0	0	81	23	0	100	1
PR.2	0	0	81	23	0	100	1
PR.3	0	0	0	1	0	100	100
PR.4	57	0	0	1	0	100	100
PR.5	0	0	0	1	0	100	100
PR.6	57	0	0	1	0	100	100
PR.7	34	12	77	32	3	100	100
ER.1	0	0	100	19	100	80	0
ER.2	0	0	100	50	0	8	100
ER.3	100	95	6	1	0	84	0
RR.1	0	0	96	2	11	100	100

Zeros are not-flagged judgments, not verified passes. Wilson intervals are provided in T04.

Table 4. Reason frequency across flag explanations

Rows 1–14 of 28

reason_id	label	Flag explanations	Share of flags
R01	Missing stated provenance	1075	33.3%
R06	Indirect capability phrasing	700	21.7%
R08	Bound wording criticized	457	14.2%
R07	Negative obligation wording	200	6.2%
R04	Notes mixed into requirement	194	6.0%
R02	Prescribed design solution	186	5.8%
R12	Environmental bounds undefined	181	5.6%
R10	Endpoint versus path accuracy	162	5.0%
R13	Particulate limit or measurement unspecified	153	4.7%
R21	Gravity versus mission-context tension	140	4.3%
R05	Multiple obligations	119	3.7%
R20	Gravity orientation or test context unclear	110	3.4%
R11	Unit or numerical equivalence	105	3.3%
R03	Abstraction or system boundary	98	3.0%

Multi-label coding: one flag can contain several reasons. These counts do not measure requirement defect prevalence.

Table 4. Reason frequency across flag explanations

Rows 15–28 of 28

reason_id	label	Flag explanations	Share of flags
R19	Energy-budget scope unclear	98	3.0%
R17	Initiating condition omitted	87	2.7%
R15	Trajectory or motion scope unspecified	80	2.5%
R18	Coordinates or reference frame omitted	61	1.9%
R23	Accuracy verification conditions missing	57	1.8%
R09	Dependence on another requirement	55	1.7%
R16	Payload or interface details omitted	35	1.1%
R24	Duration consistency questioned	16	0.5%
R28	Claimed environment-value mismatch	14	0.4%
R14	Leakage or containment undefined	13	0.4%
R27	Assumed component existence	13	0.4%
R25	Absolute prohibition unverifiable	5	0.2%
R22	Timed interval boundaries missing	4	0.1%
R26	Implied power constraint	2	0.1%

Multi-label coding: one flag can contain several reasons. These counts do not measure requirement defect prevalence.

Table 5. Same reason under two criteria in one requirement

Rows 1–15 of 15

requirement_id	reason_id	criterion_a	criterion_b	runs_both	runs_only_a	runs_only_b
ER.3	R02	A2	A9	81	19	0
ER.1	R12	A4	A9	62	38	0
PR.4	R01	A2	A9	57	0	43
PR.6	R01	A2	A9	56	0	44
ER.2	R13	A4	A5	50	50	0
FR.1	R04	A10	A6	46	43	11
ER.1	R12	A4	A5	19	81	0
PR.7	R21	A2	A9	18	16	63
PR.7	R01	A2	A9	18	1	78
PR.7	R21	A4	A9	12	4	69
PR.7	R20	A2	A4	8	4	60
PR.7	R20	A3	A4	7	5	61
ER.1	R12	A5	A9	7	12	55
PR.7	R21	A3	A9	6	1	75
PR.7	R15	A4	A5	6	11	18

Both = number of runs assigning this reason under both criteria. This association alone does not establish criterion confusion.

Table 6. Worked review cases

Rows 1–4 of 6

Record	Review focus	Question to resolve
Execution999|FR.1|A6	Notes versus singularity	The supplied PDF already labels Notes 1–4. Does the criticism concern notes, actual additional obligations, or a different input format?
Execution1006|FR.1|A4	Design freedom versus ambiguity	Note 1 intentionally leaves the path to be determined. What additional boundary would be necessary, and is the LLM imposing a design choice?
Execution999|PR.1|A9	Missing source versus incorrectness	The ConOps gives no numerical tolerance. Does this establish incorrectness, or only that correctness cannot be verified from the supplied source?
Execution999|PR.3|A10	Style-guide evidence	The explanation calls wording prohibited. Which approved style guide establishes that rule? The supplied task only gives the general criterion.

These are review prompts, not adjudicated expert findings. Full explanations remain in the linked raw-data records.

Table 6. Worked review cases

Rows 5–6 of 6

Record	Review focus	Question to resolve
Execution999|PR.7|A9	Operating versus testing context	The context names ISS and PR7 names normal gravity. Does the condition describe on-orbit use, ground testing, a loading direction, or an unsupported assumption?
Execution1002|RR.1|A4	One minute versus two motion limits	Two motions of up to 30 seconds can total one minute. Inspect whether an alleged timing contradiction is real, and what all operations includes.

These are review prompts, not adjudicated expert findings. Full explanations remain in the linked raw-data records.